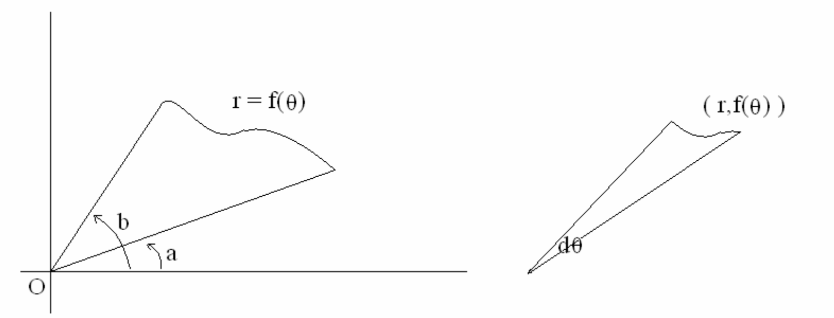
§10-4 Areas and Lengths in Polar Coordinates

Homework：5,8,11,27,31,45,55

(I) Areas



$$A=\int_a^b \frac{1}{2}r^2\,d\theta=\frac{1}{2}\int_a^b f^2(\theta)\,d\theta.$$

(II) Arc Length

$$x=f(\theta)\cos\theta \qquad y=f(\theta)\sin\theta$$

$$L=\int ds=\int\sqrt{(dx)^2+(dy)^2}=\int_a^b\sqrt{r^2+\left(\frac{dr}{d\theta}\right)^2}\,d\theta.$$

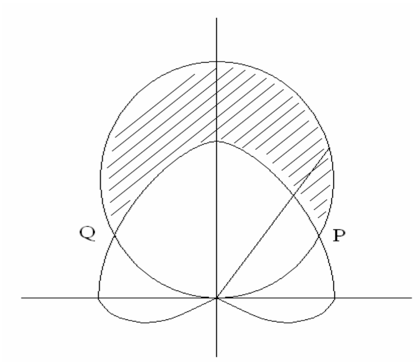
(III) Surface Area

$r=f(\theta)$, $a\le\theta\le b$, around the x-axis

$$S=\int 2\pi y\,ds=\int_a^b 2\pi\underbrace{f(\theta)\sin\theta}_{y}\underbrace{\sqrt{r^2+\left(\frac{dr}{d\theta}\right)^2}\,d\theta}_{ds}.$$

Example 1： Find the area of the region that lies inside the circle $r = 3\sin\theta$ and outside the cardioid $r = 1 + 3\sin\theta$.

Solution：



$$\begin{cases} r = 3\sin\theta \\ r = 1 + \sin\theta \end{cases} \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6} \ or \ \frac{5\pi}{6}$$

$$A = \frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \left[(3\sin\theta)^2 - (1+\sin\theta)^2 \right] d\theta$$

$$= \pi.$$

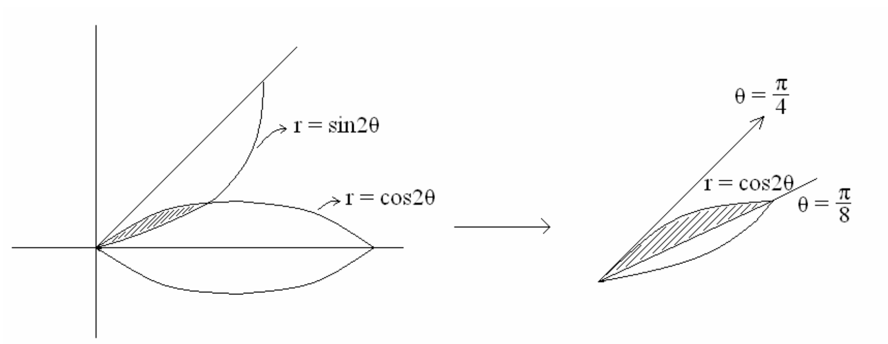
Example 2： Find the area of the region that lies inside both curves $r = \sin 2\theta$, $r = \cos 2\theta$.

Solution：

$r = \cos 2\theta$,4-leafed rose.

$$r = \sin 2\theta = \cos\left(\frac{\pi}{2} - 2\theta\right) = \cos 2\left(\frac{\pi}{4} - \theta\right) = \cos 2\left(\theta - \frac{\pi}{4}\right)$$

（將 $r = \cos 2\theta$, rotate $\frac{\pi}{4}$ 可得 $r = \sin 2\theta$ ）



$$\begin{cases} r = \sin 2\theta \\ r = \cos 2\theta \end{cases} \Rightarrow \tan 2\theta = 1 \Rightarrow \theta = \frac{\pi}{8}$$

$\Rightarrow$ 上圖斜線區域面積 $= \int_{\frac{\pi}{8}}^{\frac{\pi}{4}} \frac{1}{2}\cos^2 2\theta d\theta$

$$=\int_{\frac{\pi}{8}}^{\frac{\pi}{4}} \frac{1+\cos 4\theta}{4}\, d\theta$$

$$=-\left(\frac{\theta}{4}+\frac{\sin 4\theta}{16}\bigg|_{\frac{\pi}{8}}^{\frac{\pi}{4}}\right)$$

$$=\left(\frac{\pi}{32}-\frac{1}{16}\right)$$

$$\Rightarrow \text{總面積}=16\text{ 個斜線區域}=16\left(\frac{\pi}{32}-\frac{1}{16}\right)=\frac{\pi}{2}-1.$$

Example 3： Find the length of cardioid $r=1+\sin\theta$.

Solution：

$$L=2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{r^2+\left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

$$=2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{(1+\sin\theta)^2+\cos^2\theta}\, d\theta$$

$$=2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{2+2\sin\theta}\, d\theta$$

$$=2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\frac{4-4\sin^2\theta}{2-2\sin\theta}}\, d\theta$$

$$=4\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos\theta}{\sqrt{2-2\sin\theta}}\, d\theta \qquad u=2-2\sin\theta \;\Rightarrow\; du=-2\cos\theta d\theta$$

$$=2\int_0^4 \frac{du}{\sqrt{u}}=4u^{\frac{1}{2}}\bigg|_0^4=8.$$